

PRACTICAL COURSE WORKBOOK

Applied PostgreSQL

Model, query and explain dependable data

LEVEL	Intermediate
GUIDED TIME	5 hours across 12 lessons
PROJECT	a working data model with validated queries
SKILLS	PostgreSQL, JSONB, Triggers, Performance, Critical evaluation, Documentation
EDITION	Structured draft - version 0.9

WHAT YOU WILL BE ABLE TO DO

- Use PostgreSQL appropriately in a realistic, bounded task.
- Use JSONB appropriately in a realistic, bounded task.
- Use Triggers appropriately in a realistic, bounded task.
- Use Performance appropriately in a realistic, bounded task.

WHO THIS IS FOR

Developers and analysts who need dependable data systems.

BEFORE YOU BEGIN

- Basic programming or spreadsheet experience

Use this workbook with the online lessons. It is designed for A4 printing, handwritten evidence notes, and offline revision. It does not provide certification.

WORKBOOK GUIDE

A clear route through the course

Applied PostgreSQL is a structured, practical course covering PostgreSQL, JSONB, Triggers, Performance. It emphasizes explainable methods, checked work and a final outcome that can be reviewed.

MODULE	LESSONS AND FOCUS
01 PostgreSQL: Data modelling	1. Entities with PostgreSQL 2. Relationships with JSONB 3. Constraints with Triggers
02 JSONB: Querying	1. Selection with Performance 2. Joins with PostgreSQL 3. Aggregation with JSONB
03 Triggers: Reliability	1. Transactions with Triggers 2. Indexes with Performance 3. Backups with PostgreSQL
04 Performance: Production design	1. Security with JSONB 2. Performance with Triggers 3. Operations with Performance

HOW TO USE EACH LESSON PAGE

- 1 Read the objective and identify the decision it supports.
- 2 Attempt the retrieval prompt before checking the online explanation.
- 3 Reproduce the worked example with the stated input.
- 4 Test one normal case and one boundary or failure case.
- 5 Record the result, limitation and next action in the evidence log.

PRINT AND FILE

Print at 100% scale on A4 paper. Use double-sided printing with long-edge binding. Keep completed pages with the project files named in the evidence log.

MODULE 01

PostgreSQL: Data modelling

Apply PostgreSQL through entities, relationships, constraints.

LESSON 01 / 18 MIN

Entities with PostgreSQL

Learning objectives

- Explain entities in the context of Applied PostgreSQL.
- Apply PostgreSQL to a bounded practical task.
- Evaluate the result using explicit quality criteria.

Key terms

entities, PostgreSQL, databases

Retrieval prompt

In Applied PostgreSQL, which evidence best supports a entities result produced with PostgreSQL?

Evidence to keep

☐ Original input ☐ Observed result ☐ Boundary or failure result ☐ Limitation and next action

LESSON 02 / 22 MIN

Relationships with JSONB

Learning objectives

- Explain relationships in the context of Applied PostgreSQL.
- Apply JSONB to a bounded practical task.
- Evaluate the result using explicit quality criteria.

Key terms

relationships, JSONB, databases

Retrieval prompt

In Applied PostgreSQL, which evidence best supports a relationships result produced with JSONB?

Evidence to keep

☐ Original input ☐ Observed result ☐ Boundary or failure result ☐ Limitation and next action

LESSON 03 / 26 MIN

Constraints with Triggers

Learning objectives

- Explain constraints in the context of Applied PostgreSQL.
- Apply Triggers to a bounded practical task.
- Evaluate the result using explicit quality criteria.

Key terms

constraints, Triggers, databases

Retrieval prompt

In Applied PostgreSQL, which evidence best supports a constraints result produced with Triggers?

Evidence to keep

[] Original input [] Observed result [] Boundary or failure result [] Limitation and next action

MODULE REVIEW

Turn the module into usable evidence

1. RETRIEVE

Without checking the lessons, explain the module's three most important ideas in your own words.

2. APPLY

Describe the example or file you produced, the input used, and the result you observed.

3. REVIEW

Record one uncertainty, one source to recheck, and the next deliberate practice action.

MODULE 02

JSONB: Querying

Apply JSONB through selection, joins, aggregation.

LESSON 04 / 30 MIN

Selection with Performance

Learning objectives

- Explain selection in the context of Applied PostgreSQL.
- Apply Performance to a bounded practical task.
- Evaluate the result using explicit quality criteria.

Key terms

selection, Performance, databases

Retrieval prompt

In Applied PostgreSQL, which evidence best supports a selection result produced with Performance?

Evidence to keep

[] Original input [] Observed result [] Boundary or failure result [] Limitation and next action

LESSON 05 / 18 MIN

Joins with PostgreSQL

Learning objectives

- Explain joins in the context of Applied PostgreSQL.
- Apply PostgreSQL to a bounded practical task.
- Evaluate the result using explicit quality criteria.

Key terms

joins, PostgreSQL, databases

Retrieval prompt

In Applied PostgreSQL, which evidence best supports a joins result produced with PostgreSQL?

Evidence to keep

[] Original input [] Observed result [] Boundary or failure result [] Limitation and next action

LESSON 06 / 22 MIN

Aggregation with JSONB

Learning objectives

- Explain aggregation in the context of Applied PostgreSQL.
- Apply JSONB to a bounded practical task.
- Evaluate the result using explicit quality criteria.

Key terms

aggregation, JSONB, databases

Retrieval prompt

In Applied PostgreSQL, which evidence best supports a aggregation result produced with JSONB?

Evidence to keep

[] Original input [] Observed result [] Boundary or failure result [] Limitation and next action

MODULE REVIEW

Turn the module into usable evidence

1. RETRIEVE

Without checking the lessons, explain the module's three most important ideas in your own words.

2. APPLY

Describe the example or file you produced, the input used, and the result you observed.

3. REVIEW

Record one uncertainty, one source to recheck, and the next deliberate practice action.

MODULE 03

Triggers: Reliability

Apply Triggers through transactions, indexes, backups.

LESSON 07 / 26 MIN

Transactions with Triggers

Learning objectives

- Explain transactions in the context of Applied PostgreSQL.
- Apply Triggers to a bounded practical task.
- Evaluate the result using explicit quality criteria.

Key terms

transactions, Triggers, databases

Retrieval prompt

In Applied PostgreSQL, which evidence best supports a transactions result produced with Triggers?

Evidence to keep

☐ Original input ☐ Observed result ☐ Boundary or failure result ☐ Limitation and next action

LESSON 08 / 30 MIN

Indexes with Performance

Learning objectives

- Explain indexes in the context of Applied PostgreSQL.
- Apply Performance to a bounded practical task.
- Evaluate the result using explicit quality criteria.

Key terms

indexes, Performance, databases

Retrieval prompt

In Applied PostgreSQL, which evidence best supports a indexes result produced with Performance?

Evidence to keep

☐ Original input ☐ Observed result ☐ Boundary or failure result ☐ Limitation and next action

LESSON 09 / 18 MIN

Backups with PostgreSQL

Learning objectives

- Explain backups in the context of Applied PostgreSQL.
- Apply PostgreSQL to a bounded practical task.
- Evaluate the result using explicit quality criteria.

Key terms

backups, PostgreSQL, databases

Retrieval prompt

In Applied PostgreSQL, which evidence best supports a backups result produced with PostgreSQL?

Evidence to keep

[] Original input [] Observed result [] Boundary or failure result [] Limitation and next action

MODULE REVIEW

Turn the module into usable evidence

1. RETRIEVE

Without checking the lessons, explain the module's three most important ideas in your own words.

2. APPLY

Describe the example or file you produced, the input used, and the result you observed.

3. REVIEW

Record one uncertainty, one source to recheck, and the next deliberate practice action.

MODULE 04

Performance: Production design

Apply Performance through security, performance, operations.

LESSON 10 / 22 MIN

Security with JSONB

Learning objectives

- Explain security in the context of Applied PostgreSQL.
- Apply JSONB to a bounded practical task.
- Evaluate the result using explicit quality criteria.

Key terms

security, JSONB, databases

Retrieval prompt

In Applied PostgreSQL, which evidence best supports a security result produced with JSONB?

Evidence to keep

☐ Original input ☐ Observed result ☐ Boundary or failure result ☐ Limitation and next action

LESSON 11 / 26 MIN

Performance with Triggers

Learning objectives

- Explain performance in the context of Applied PostgreSQL.
- Apply Triggers to a bounded practical task.
- Evaluate the result using explicit quality criteria.

Key terms

performance, Triggers, databases

Retrieval prompt

In Applied PostgreSQL, which evidence best supports a performance result produced with Triggers?

Evidence to keep

☐ Original input ☐ Observed result ☐ Boundary or failure result ☐ Limitation and next action

LESSON 12 / 30 MIN

Operations with Performance

Learning objectives

- Explain operations in the context of Applied PostgreSQL.
- Apply Performance to a bounded practical task.
- Evaluate the result using explicit quality criteria.

Key terms

operations, Performance, databases

Retrieval prompt

In Applied PostgreSQL, which evidence best supports a operations result produced with Performance?

Evidence to keep

[] Original input [] Observed result [] Boundary or failure result [] Limitation and next action

MODULE REVIEW

Turn the module into usable evidence

1. RETRIEVE

Without checking the lessons, explain the module's three most important ideas in your own words.

2. APPLY

Describe the example or file you produced, the input used, and the result you observed.

3. REVIEW

Record one uncertainty, one source to recheck, and the next deliberate practice action.

CAPSTONE PROJECT

Produce a reviewable Applied PostgreSQL project using PostgreSQL, JSONB, Triggers.

DELIVERABLE	A working Applied PostgreSQL artefact plus an evidence-based self-review.
TOOLS	A suitable PostgreSQL environment, A plain-text decision log, Test data or realistic sample material

Production plan

- 1 Define the intended user, outcome and constraints.
- 2 Create the smallest complete result using PostgreSQL.
- 3 Test one normal case, one boundary case and one failure response.
- 4 Revise the work from the evidence and preserve before-and-after results.
- 5 Prepare a concise handover containing method, limitations and next step.

Definition of done

- The output matches the stated outcome.
- Inputs and decisions are reproducible.
- Boundary and failure evidence is included.
- Limitations and responsibility considerations are explicit.

PROJECT EVIDENCE

Document decisions another person can review

1. PURPOSE AND USER

State the intended user, need, expected outcome and constraints.

2. INPUTS AND ASSUMPTIONS

List source files, data, versions, permissions and assumptions.

PROJECT EVIDENCE

Tests, limitations and revision

3. NORMAL CASE

Expected result, observed result and evidence location.

4. BOUNDARY OR FAILURE CASE

What was difficult, what happened and why it matters.

5. REVISION AND LIMITATION

What changed, what improved and what remains unproven.

REVIEW AND SOURCES

Make the work credible and repeatable

Final self-review

- Can another learner repeat the method?
- Did I test a difficult case?
- Did I avoid unsupported claims?
- Is the next action proportionate to the remaining risk?

FINAL DECISION

Is the project ready to share? State the evidence, remaining risk, owner and next review date.

Authoritative references

PostgreSQL Tutorial

PostgreSQL Global Development Group - reviewed 2026-08-21

<https://www.postgresql.org/docs/current/tutorial.html>

SQL Language

PostgreSQL Global Development Group - reviewed 2026-08-21

<https://www.postgresql.org/docs/current/sql.html>

SOURCE NOTE

Product versions, laws and professional standards can change. Recheck the linked primary source before applying version-sensitive information.

NEXT IMPROVEMENT

Choose one weakness found during review and improve it before extending the PostgreSQL scope.